

Code	Course Title	CHs	Pre-Req
CS617	Advanced Operating Systems	3(3+0)	None
Recommended Texts			
- Distributed Systems: Principles and Paradigms, Andrew S. Tanenbaum, Maarten van Steen , CreateSpace Independent Publishing Platform; 2nd edition (February 26, 2016) , ISBN-13 : 978-1530281756 - Modern Operating Systems, Tanenbaum, Andrew S. Tanenbaum, Pearson India; 4th edition (March 25, 2016), ISBN-13 : 978-9332575776 - At least 5 research papers from Scopus indexed international conferences - At least 5 research papers from journals papers indexed in Q1 or Q2 WoS			
Course Description			
The course will focus on the engineering and performance trade-offs in the design of operating systems. The purpose will be to teach not only what operating systems are and how they work today, but also why they are designed the way they are and how they are likely to evolve in the future. The emphasize will be on the practical aspects of the topics through the case study of Linux kernel as an example of a commercial operating system.			
Course Objectives			
- Understand in great detail how and why different parts of an operating system work. - Understand the engineering tradeoffs involved in the design of various sub-modules of an operating system. - Understand how operating systems are structured, what are alternative OS architectures and how different modules interact together to form a cohesive and complex system.			
Week Wise Topics to be Covered			
W1	Review of Operating Systems concepts	W9	Synchronization in distributed systems
W2	Hardware concepts of distributed systems	W10	Clock synchronization
W3	Software concepts and design issues	W11	Mutual exclusion; Election algorithms
W4	Communication in distributed systems	W12	Transaction and concurrent control
W5	Threads and thread usage Multithreading operating system	W13	Deadlock in distributed systems
W6	Client – server model Implementation of Client-server model	W14	Processor Allocation
W7	Remote procedure call	W15	Case Studies / Projects / Demo
W8	Implementation of remote procedure call	W16	Presentations / Peer Review / Q&A Sessions / Discussions
MID EXAM/ MID OF SEMESTER		Final EXAM	

